CivicResolve: An AI-Based Public Grievance Management System

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***Abstract*—Public grievance handling suffers from incomplete submissions, inconsistent prioritization, incorrect departmental routing, and limited transparency. This paper presents CivicResolve, a web-based grievance management system that uses symbolic AI as a decision-support layer. A Constraint Satisfaction Problem (CSP) component validates structured complaint data; a Forward Chaining production rule engine derives department, priority, and escalation recommendations; a precedence-based conflict-resolution mechanism handles competing priorities; and a deterministic explanation engine reports the facts and rules behind each recommendation. Citizens, administrators, and department officers work through role-based access and a defined complaint lifecycle, with administrators retaining authority over assignment, overrides, and closure. The design avoids machine learning, natural language processing, and large language models, favouring traceability and auditability. A test plan covering validation, inference, conflict handling, access control, and workflow integrity is proposed.**

***Index Terms*—public grievance management, constraint satisfaction, forward chaining, production rules, conflict resolution, explainable AI, e-governance.**

I. Introduction

Public grievance management lets citizens report problems with civic services such as roads, waste, streetlights, water supply, and drainage, and obtain action from the responsible authorities. In poorly integrated processes, citizens struggle to identify the correct department or follow progress, while officials spend effort checking incomplete submissions, judging urgency, and forwarding cases. Opaque decisions further reduce transparency and public confidence [1], [5].

A digital portal supports registration and tracking, but does not by itself guarantee consistent validation, systematic prioritization, or explainable recommendations. Explanations influence how affected people perceive algorithmic decisions in government [3], and officials may over-rely on or selectively follow automated advice [6]. These findings motivate decision support that is transparent and leaves final responsibility with humans.

This paper presents CivicResolve, which combines a grievance workflow with symbolic AI: CSP validation, Forward Chaining inference, explicit conflict resolution, and a deterministic explanation engine. The main contributions are: (i) a validation-before-inference pipeline for structured complaints; (ii) rule-based recommendations of department, priority, and escalation; (iii) a documented precedence policy for competing recommendations; (iv) explanations generated from the actual inference trace; and (v) a human-supervised lifecycle with recorded overrides.

II. Related Work

Srivatsa et al. study digital grievance redressal in India from a participatory-governance view, based on a survey of 157 urban residents and stakeholder interviews, and stress that citizens should be able to understand and take part in the process [1]. Infra-Alert is a web platform for reporting infrastructure problems with image upload, geolocation, departmental routing, and status tracking on a MySQL back end [2]. Kumar et al. surveyed 677 participants and relate service quality factors, including security, privacy, reliability, and information quality, to satisfaction with grievance portals [5].

On explainability, Aoki et al. show that the type of explanation affects perceived accuracy, fairness, and trustworthiness of public-sector algorithmic decisions [3], and Papadakis et al. discuss transparency, technical challenges, and regulatory considerations in AI-supported policymaking [4]. Alon-Barkat and Busuioc examine automation bias and selective adherence among public decision-makers [6]. These works address portals, participation, quality, and trust, but do not describe the specific combination of CSP validation, Forward Chaining, precedence-based conflict resolution, and deterministic explanations in one supervised grievance workflow, which is the gap CivicResolve targets. Unlike learning-based approaches, it applies a defined knowledge base to structured facts.

III. Proposed System

A. Architecture

Citizens submit a structured complaint. The system validates it, derives recommendations, resolves conflicts, records an explanation, and presents the result to an administrator, who assigns the case. Officers process it and update status; the administrator verifies the outcome and closes it; citizens track progress throughout (Fig. 1). The system is organized into seven layers: presentation; complaint management; CSP validation; knowledge base and inference; conflict resolution; explanation and audit; and data storage. Validation is kept separate from recommendation so that invalid or inconsistent input is never treated as verified fact, and inference is kept separate from administrative authority.

B. CSP Validation

A CSP is defined by variables, domains, and constraints over their combinations. Here the variables are complaint fields such as category, location, severity, and urgency. Constraints require mandatory fields to be present, values to belong to permitted sets, and any defined cross-field conditions to hold. If validation fails, the system reports the specific issue and requests correction or review; it does not silently alter the citizen's answers. Only validated facts reach the inference engine. Validation checks consistency with defined constraints and does not prove that a claim is true in the real world.

C. Forward Chaining Inference

Forward Chaining is data-driven: starting from known facts, it repeatedly fires applicable IF–THEN rules, adding derived facts to working memory until no new fact can be derived or a stopping condition is met. The knowledge base holds production rules whose conditions are matched against working memory. Illustrative rules are:

- *R1:* IF the complaint is validated AND severity is high AND urgency conditions hold THEN recommend HIGH priority.
- *R2:* IF the complaint is validated AND a predefined critical-risk condition holds THEN recommend CRITICAL priority.
- *R3:* IF the validated category matches a department mapping THEN recommend the mapped department.

These show the method; the deployed system uses the rule definitions in its own knowledge base. The engine records each rule that fires together with its supporting facts.

D. Conflict Resolution

Rules may yield competing priority recommendations. CivicResolve applies the explicit precedence CRITICAL > HIGH > MEDIUM > LOW and preserves the rules behind every competing recommendation in the trace. Precedence is not a substitute for validation: if the input facts violate a consistency constraint, the case is flagged for administrative review rather than resolved by choosing the highest priority. Conflicting department recommendations are handled by a separately configured department-resolution policy or referred to an administrator.

E. Deterministic Explanation Engine

The explanation engine converts the inference trace into template-based text. It records the validated facts used, rules fired, priority and department recommendations, the conflict-resolution result, any validation issue or unresolved conflict, and any administrator override with its reason. For example: "The complaint was recommended as HIGH priority because the validated severity and urgency facts satisfied the applicable high-priority rule." Wording is generated from rules that actually fired, so output is repeatable and auditable without a language model, provided the knowledge base and trace are correctly maintained.

F. Human-Supervised Decision-Making

Recommendations are advisory. Administrators review them, assign complaints, monitor progress, and verify resolution before closure. An override records the original recommendation, the revised decision, the authorized user, and the reason. Officers act only on complaints assigned to them. This separation of duties addresses the over-reliance risk identified in [6] and keeps the lifecycle auditable.

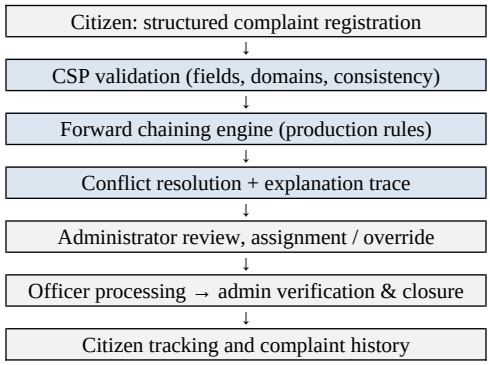


Fig. 1. CivicResolve processing pipeline (shaded blue: symbolic AI layer).

IV. Modules

The system comprises six modules. *Authentication and role management* enforces Citizen, Administrator, and Officer permissions. *Complaint registration and CSP validation* provides the structured form and stores each validation outcome. *Knowledge base and Forward Chaining engine* holds the rules and derives department, priority, and escalation recommendations; rule changes should be documented and tested for unintended conflicts. *Conflict resolution and explanation* applies precedence and reports issues rather than inventing conclusions when input conflicts prevent reliable inference. *Assignment and officer management* links recommendations to operational responsibility and records permitted overrides. *Tracking, notifications, and dashboard* maintains the lifecycle, timestamps, and status history; dashboard statistics are computed from stored records.

V. Implementation and Evaluation Plan

The implementation comprises a web interface for the three roles, structured forms, a complaint data store (users, complaints, assignments, status history, audit records), the CSP validator, the production-rule knowledge base and Forward Chaining engine, the conflict-resolution and explanation components, and workflow and access-control mechanisms. Platform-specific details (forms, workflows, scripting functions, or any separate inference back end) should be reported exactly as built; unimplemented components must not be presented as completed.

Table I lists a representative subset of the proposed test plan, which covers valid, invalid, urgent, conflicting, and incomplete complaints, as well as role-based and workflow tests. The outcomes are expected behaviours, not reported results.

TABLE I. PROPOSED TEST CASES (SUBSET)

ID	Scenario	Expected outcome
TC01	All required fields valid	Passes validation; proceeds to inference
TC02	Mandatory field omitted	Missing field flagged; processing blocked
TC03	Value outside allowed domain	Value rejected or flagged

ID	Scenario	Expected outcome
TC04	High-priority rule satisfied	Rule-specified recommendation produced
TC06	Multiple priority recommendations	Precedence applied; result recorded
TC07	Consistency constraint violated	Conflict flagged; routed for review
TC09	Explanation generated	Matches facts and rules in trace
TC10	Citizen attempts admin action	Access denied
TC12	Administrator override	Change and reason in audit history
TC13	Invalid status transition	Rejected per workflow rules
TC14	Closure without verification	Closure prevented

Evaluation should cover input validation (missing values, invalid domains, violated constraints), rule behaviour (expected recommendations, only intended rules firing), consistent precedence handling, agreement between explanations and traces, role-based access, workflow integrity, and recorded overrides. A completed study should report the number of cases executed, passed, and failed, with defects and corrective actions. Accuracy, response time, or success-rate figures should be reported only when supported by measurement data; none are claimed here.

Limitations

Recommendations depend on the coverage and correctness of the predefined constraints and rules; complaints outside the knowledge base may need additional rules or administrative review. CSP validation cannot verify real-world truth, and a deterministic engine does not learn from historical data. Explanation quality depends on an accurate trace and complete templates, and the system relies on correct permissions, reliable storage, and properly configured workflows.

VI. Conclusion and Future Work

CivicResolve combines CSP validation, Forward Chaining inference, explicit conflict resolution, and deterministic explanations within a role-based grievance workflow. Its recommendations are traceable to explicit rules and remain subject to human review, with permitted overrides recorded for accountability. Practical effectiveness must still be established through testing with representative complaint cases.

Future work includes expanding the knowledge base; a controlled interface for rule review, versioning, and testing; procedures for department conflicts, missing facts, and ambiguous cases; richer audit reports; multilingual and accessible interfaces; stronger security and privacy controls; larger measured evaluations; verified duplicate-complaint handling; and secure integration with approved public-service systems.

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